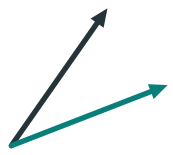
01

Bring them together.

They match.

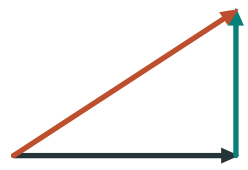


Turn one. Compare again.

02

Take this step.

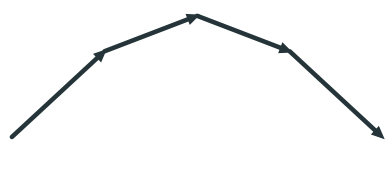
Then take this step.



One step reaches the same end.

03

A few steps.



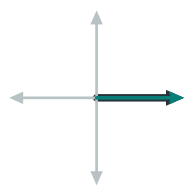
More, smaller steps.

Follow the change more closely.

04

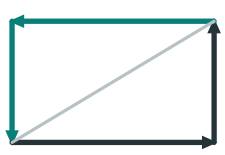
Start together.

Turn just one.

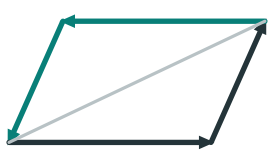


Keep turning the same way.
It comes back.

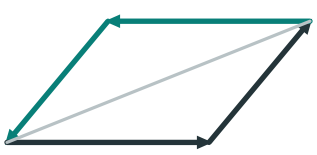
05



Two pieces make this patch.

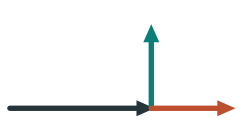


Slide the top sideways.



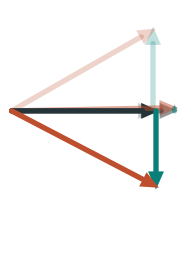
Same area. Different shape.

06



Now, black changes
at green’s pace.

At that same instant, green changes
at black’s pace, the opposite way.

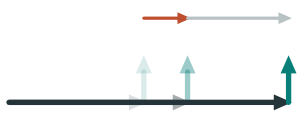


Under both rules,
the joined arrow turns.

07

New rule:
green sets the slide.

Shift, then slide.



Slide, then shift.
Same end.

08

Black advances.

Green stays the same.

Every step keeps that amount.

Mathematics and assumptions

Companion to strips 01–08 · supplied structure, derived consequences, explicit limits

What these pictures can establish

A proof sketch for a supplied classical model with one degree of freedom: local Hamiltonian structure and a Noether example. This is not a derivation of mechanics from marks alone, nor of quantum mechanics or the supplied alpha paper.

01–04 · Comparing, composing, refining, returning

We supply an affine plane, comparison by translation, and reversible rotations. Tip-to-tail composition gives vector addition. Strip 3 samples a prescribed differentiable path; drawings alone do not prove a smooth limit. Strip 4 introduces cyclic position (angular phase), not the specialized geometric phase of transport/holonomy, and not a quantum amplitude. It shows three intermediate directions and a return.

05 · The extra structure that makes Hamilton possible

We now supply a STATE plane with dimensionless conjugate coordinates q,p and oriented area $\omega = dq \wedge dp$. In strips 6–8 black measures q (right positive), green measures p (up positive); left/down are negative. The green tip locates the state relative to the black tail. These are state components, not spatial directions. Conjugacy is assumed, not established by a right angle. In physical units p dq has units of action. Momentum need not equal velocity.

05 → 06 · The local derivation

Let u,v be rates: a dot means change per unit time. Assume an autonomous (time-independent), C^1 (continuously differentiable) rule preserving oriented area for EVERY small cell. Local expansion must cancel: $\partial u/\partial q + \partial v/\partial p = 0$. A partial derivative is a slope holding the other coordinate fixed. This lets one function H generate both rates. Formally, $\alpha = u\,dp - v\,dq$ is closed: its integral around any sufficiently small closed loop vanishes. On a simply connected patch (no holes), integrating α is path-independent and gives H. Thus

$$\dot{q} = u = \partial H/\partial p, \qquad \dot{p} = v = -\partial H/\partial q.$$

H is fixed up to a constant. Strip 5 illustrates the area hypothesis; it does not prove it for every rule. Topology can obstruct a global H. Physical energy needs a physical model.

06 · An exact turning rule

Choose $H = (q^2+p^2)/2$. The two rate panels use the SAME state (2.1,1.2), at the SAME instant. Orange lengths are 1.2 and 2.1 on a common rate scale. They are not successive kicks. The last panel joins each pair as in strip 2: three exact states at times 0, 0.5, 1 share one origin and scale; pale means earlier. The exact evolution is

$$q' = q\cos h + p\sin h, \qquad p' = p\cos h - q\sin h.$$

This preserves ω and H. The Euler assignment $(q + h\,p, p - h\,q)$ multiplies area and q^2+p^2 by $1+h^2$ instead. Circles belong to this chosen example and scaling, not every Hamiltonian system.

07 · A different rule and its continuous symmetry

Switch to $H = p^2/2$: $\Phi_t(q,p) = (q+t\,p, p)$. The shift $T_a(q,p) = (q+a,p)$ preserves the specified area (is canonical) and H, so $\Phi_t T_a = T_a \Phi_t$. Both orders start at (2,0.65), use shift a=1.5 and time t=1, and end at (4.15,0.65). Pale intermediate q values are 3.5 and 2.65. Grey always means shift; orange means slide. Strip 8 repeats the unshifted run at times 0,1,2. The oscillator does NOT have this translation symmetry.

08 · The Noether charge and why it is conserved

Translation is generated by $Y = \partial/\partial q$; its pairing with ω gives dp, so the charge is $J = p$. Since H has no q dependence, $dJ/dt = \dot{p} = -\partial H/\partial q = 0$. Equivalently, use $S = \int (p\,\dot{q} - H)dt$. Independent interior variations of q,p, with $\delta q = 0$ at time endpoints, give Hamilton's equations. Constant q translations preserve the action and map endpoint data to translated data; they are not fixed-endpoint variations. Noether conservation holds along stationary trajectories. Equal drawn marks alone do not prove this theorem.

Limits and the remaining debt

Supplied: geometry, oriented pairing, regularity, the two model rules and variational hypotheses. Derived conditionally: local Hamiltonian form, exact example flows and conserved translation charge. Not derived: the earlier Genesis/L1 result, an identity of holonomy with Noether, quantization, Born weights, Everett branching or Bell correlations. Novice comprehension remains to be tested.